# Nonlinear SORh and SORp Parameter Studies

Omega calibration, timestep selection, CFL decomposition, coefficient degeneracy, initial-guess sensitivity, and optimized grid scaling

**Primary result: optimized SORh becomes increasingly faster than optimized SORp as the grid is refined, reaching a 5.29x sweep advantage at N=64. The coefficient-degeneracy study shows that this advantage is specific to the nonlinear degenerate regime: it shrinks continuously and reverses for the constant-coefficient control.**

## Executive findings

- The experimental SORh optima remain close to the linear-theory formula, while the degenerate nonlinear SORp optima shift substantially upward and sit near sharp stability boundaries.
- Adaptive timesteps are neutral for SORh but increase SORp sweeps by 6-8% and roughly double SORp elapsed time.
- Holding omega x CFL fixed does not preserve SORh convergence. At low CFL it forces omega above the stable range; at high CFL it reduces directional propagation and raises the sweep count.
- SORh is about 2.9-3.0x faster through the strongly degenerate cases, but constant-coefficient SORp is 11% faster than SORh.
- The linear initial guess u0=x reduces SORh sweeps by 28-32%, but the zero-initial-state SORp optima stagnate from that different initial state.
- Both methods converge to solutions agreeing within approximately 1.0e-6, so the performance differences are not caused by different final solutions.

Prepared from completed deterministic study outputs on August 3, 2026. Sweep counts are the primary performance measure; wall-clock time is reported as supporting evidence.

# 1. Experimental setup

The test problem is the two-dimensional nonlinear diffusion equation div(v(u) grad(u)) = 0 on the unit square. The baseline coefficient is v(u)=u/(1+u), which vanishes at u=0 and is therefore elliptically degenerate. Dirichlet boundary data are u=x. All reported runs use lexicographic ordering, CFL=0.8, and a zero initial guess unless a study explicitly changes one of those choices.

The solvers are the hyperbolic relaxation method SORh and the parabolic relaxation method SORp. A run is converged only when both the normalized update and nonlinear residual are at most 1.0e-5. Exploratory searches are capped at 20,000 sweeps and include residual-growth and stagnation safeguards. Omega selection uses the minimum-sweep point on the first contiguous converged branch, excluding isolated stable pockets beyond the first failure.

# 2. Omega calibration

The linear references are the formulas already implemented in the configuration object: omega_H,ref = 1/(1+sqrt(1-c_max^2)) and omega_P,ref = 2/[CFL(1+sqrt(1-c_max^2))]. The nonlinear calibration sweeps omega around these references and refines the best stable interval.

| N | SORh omega | H formula | H sweeps | SORp omega | P formula | P sweeps |
|---|---|---|---|---|---|---|
| 16 | 0.8323 | 0.8368 | 139 | 2.7300 | 2.0919 | 221 |
| 32 | 0.8950 | 0.9107 | 295 | 2.6800 | 2.2768 | 863 |
| 64 | 0.9368 | 0.9532 | 662 | 2.6150 | 2.3831 | 3502 |

**Table 1.** *Selected nonlinear optima compared with the linear-theory reference values.*

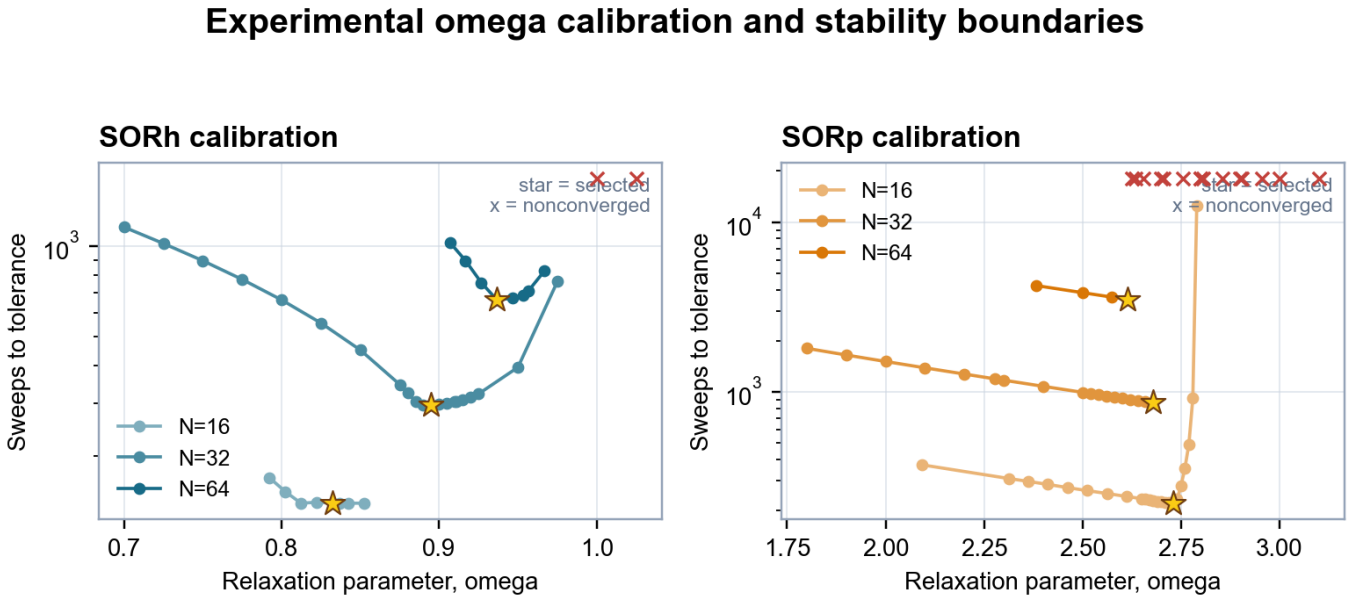



**Figure 1.** *Calibration curves for all three grids. Yellow stars mark selected optima. Red x symbols mark nonconverged candidates at a common display height rather than their stopping sweep count.*

## Calibration findings

- SORh remains close to the linear reference: the selected values differ by -0.5%, -1.7%, and -1.7% for N=16, 32, and 64.
- SORp shifts well above the linear reference in the degenerate nonlinear problem: +30.5% at N=16, +17.7% at N=32, and +9.7% at N=64.
- The SORp optimum is controlled by a sharp stability transition. At N=16 the sweep count falls to 221 at omega=2.73, then rises rapidly before stagnation at 2.80. At N=64, omega=2.615 converges in 3,502 sweeps while 2.625 stagnates.
- The calibrated SORp value therefore cannot be obtained reliably from the linear formula alone; direct nonlinear stability testing is necessary.

**Interpretation: coefficient degeneracy creates a strong nonlinear shift in the useful SORp relaxation range, whereas SORh retains a near-formula optimum. This difference anticipates the degeneracy and grid-scaling results below.**

# 3. Fixed versus adaptive timestep

For each method and grid, the fixed-timestep run uses the calibrated omega. The adaptive study then performs a local omega search because changing the timestep rule can move the stability boundary. The table reports the best converged adaptive candidate rather than forcing the fixed-timestep omega.

| N | Method | Fixed omega / sweeps | Adaptive omega / sweeps | Sweep change | Time ratio |
|---|---|---|---|---|---|
| 32 | SORh | 0.8950 / 295 | 0.8950 / 293 | -0.7% | 0.99x |
| 32 | SORp | 2.6800 / 863 | 2.6300 / 918 | +6.4% | 2.09x |
| 64 | SORh | 0.9368 / 662 | 0.9368 / 660 | -0.3% | 1.00x |
| 64 | SORp | 2.6150 / 3502 | 2.5150 / 3771 | +7.7% | 2.09x |

**Table 2.** *Best fixed and adaptive cases after local omega calibration.*

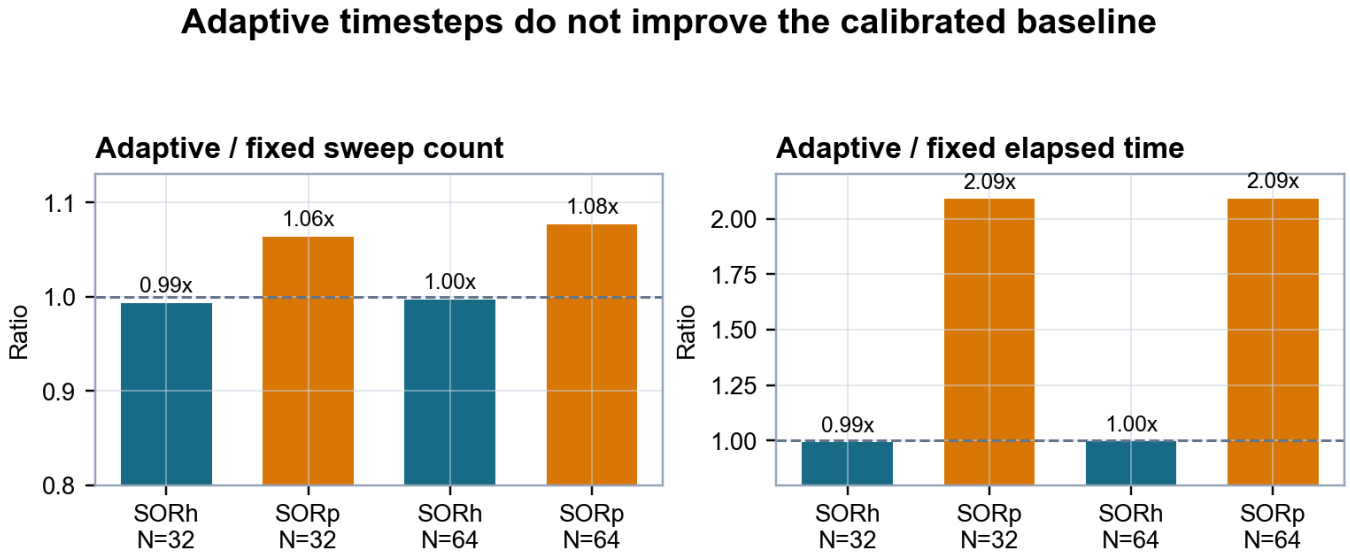



**Figure 2.** *Ratios below one favor the adaptive timestep. Adaptive SORh is essentially unchanged, while adaptive SORp requires more sweeps and about twice the elapsed time.*

- SORh changes by only -0.7% and -0.3% in sweep count at N=32 and N=64. Its adaptive mean timestep returns close to the fixed value after a larger initial step.
- SORp requires lower adaptive optima, 2.63 at N=32 and 2.515 at N=64, but still needs 6.4% and 7.7% more sweeps.
- Adaptive SORp elapsed time is 2.09x the fixed baseline on both grids, indicating substantial per-sweep overhead in addition to the larger sweep count.
- Several higher adaptive SORp candidates become nonfinite or exceed the residual guard. Fixed timesteps are therefore both faster and more robust for this test problem.

# 4. Separating omega and CFL in SORh

This study compares two paths through parameter space at N=32. The fixed-omega path holds omega_H=0.895 while CFL varies from 0.5 to 1.0. The fixed-product path holds omega_H x CFL=0.716, so omega increases when CFL decreases and decreases when CFL increases. If only the residual-injection product mattered, the second path would have similar behavior at every CFL.

| CFL | Fixed omega | Sweeps | Fixed product omega | Result |
|---|---|---|---|---|
| 0.5 | 0.895 | 467 | 1.432 | Failed |
| 0.6 | 0.895 | 391 | 1.193 | Failed |
| 0.7 | 0.895 | 336 | 1.023 | Failed |
| 0.8 | 0.895 | 295 | 0.895 | 295 |
| 0.9 | 0.895 | 263 | 0.796 | 606 |
| 1.0 | 0.895 | 238 | 0.716 | 856 |

**Table 3.** *SORh sweep counts along fixed-omega and fixed omega x CFL paths.*

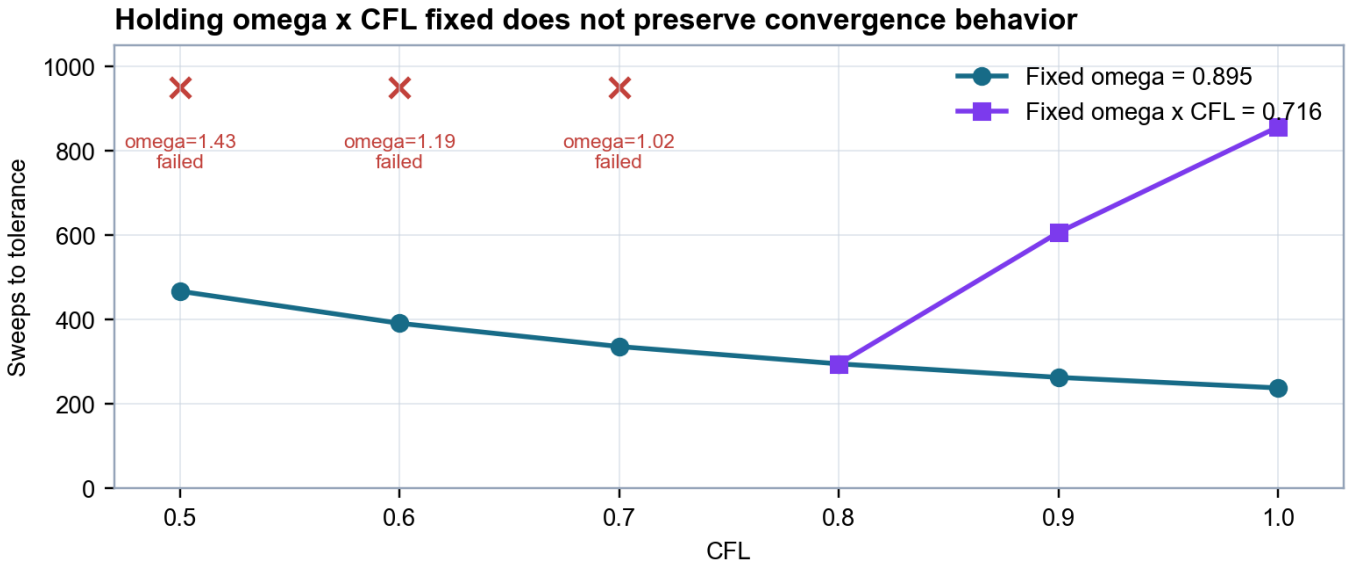



**Figure 3.** *The fixed-product path fails for CFL=0.5-0.7 and becomes progressively slower for CFL=0.9-1.0. The fixed-omega path remains stable and accelerates monotonically.*

- At fixed omega, increasing CFL strengthens the residual correction and reduces the sweep count from 467 to 238 without changing the directional relaxation parameter.
- At fixed omega x CFL, low CFL forces omega above one and causes immediate or rapid divergence. High CFL forces a smaller omega and raises the sweep count to 606 and 856.
- The result directly rejects a one-parameter description based only on omega x CFL. The propagation role of omega remains independently important.

# 5. Coefficient-degeneracy study

The regularized family uses a normalized coefficient proportional to epsilon + u/(1+u), with v(1)=0.5 held fixed. Its ellipticity ratio is v_min/v_max = epsilon/(0.5+epsilon). The sequence ranges from the original degenerate coefficient at ratio zero to ratios 0.002, 0.020, 0.167, and 0.667. A constant v=0.5 case provides the nondegenerate linear control. Each coefficient and method receives a local omega search.

| Coefficient | Ratio | H omega | H sweeps | P omega | P sweeps | P/H |
|---|---|---|---|---|---|---|
| epsilon=0 | 0.000 | 0.895 | 295 | 2.680 | 863 | 2.93x |
| epsilon=0.001 | 0.002 | 0.895 | 294 | 2.679 | 860 | 2.93x |
| epsilon=0.01 | 0.020 | 0.895 | 287 | 2.672 | 841 | 2.93x |
| epsilon=0.1 | 0.167 | 0.913 | 232 | 2.613 | 694 | 2.99x |
| epsilon=1 | 0.667 | 0.920 | 145 | 2.511 | 230 | 1.59x |
| constant | 1.000 | 0.926 | 123 | 2.277 | 109 | 0.89x |

**Table 4.** *Locally optimized performance as coefficient degeneracy is removed.*

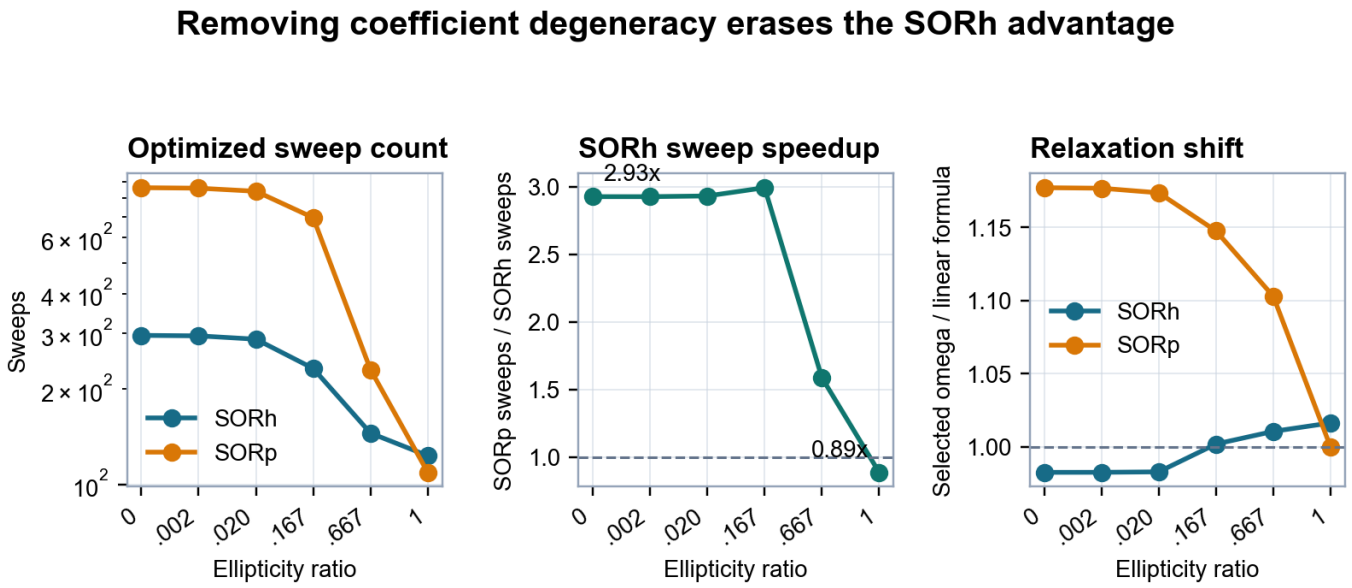



**Figure 4.** *Degeneracy controls both the sweep advantage and the SORp omega shift. The middle panel reproduces the SORh speedup diagnostic; the right panel normalizes selected omega by the linear formula.*

## Degeneracy findings

- For ratios from 0 through 0.167, SORh is consistently 2.93-2.99x faster in sweep count. Mild regularization does not remove the advantage.
- At ratio 0.667 the advantage falls to 1.59x. In the constant-coefficient control, SORp takes 109 sweeps versus 123 for SORh, so SORp is about 11% faster.
- SORp omega decreases from 2.68 in the degenerate problem to the linear formula value 2.2768 in the constant control. SORh omega changes much less, from 0.895 to 0.926.
- This is the strongest evidence in the study suite that SORh's benefit is not a generic per-sweep implementation advantage. It is associated with the nonlinear, spatially varying, near-degenerate coefficient.

**Interpretation: SORh appears to retain effective directional propagation when the local diffusion coefficient becomes very small. SORp recovers and ultimately wins only as the operator approaches the uniformly elliptic constant-coefficient limit. This supports, but does not by itself prove, the proposed directional-propagation explanation.**

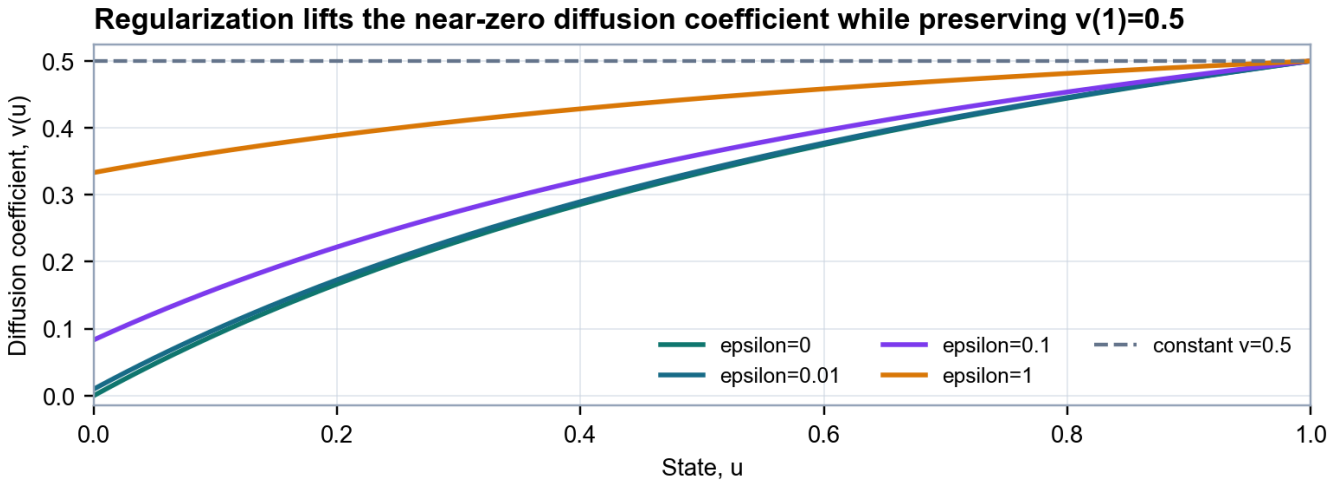



**Figure 5.** *Normalized coefficient profiles used in the regularization study. All nonconstant curves satisfy v(1)=0.5; increasing epsilon lifts v(0) and removes the degenerate endpoint.*

# 6. Initial-guess sensitivity

The baseline zero state is compared with u0=x while retaining the omega values calibrated from the zero initial state. This is intentionally a robustness test rather than a new omega calibration for each initial condition.

| N | Method | Zero guess | u0=x | u0=x best residual | Outcome |
|---|---|---|---|---|---|
| 32 | SORh | 295 | 201 | 9.79e-06 | Converged |
| 32 | SORp | 863 | 2500 | 5.04e-05 | Stagnated |
| 64 | SORh | 662 | 475 | 9.97e-06 | Converged |
| 64 | SORp | 3502 | 3000 | 3.29e-04 | Stagnated |

**Table 5.** *Sensitivity to the initial state at the zero-state calibrated omega.*

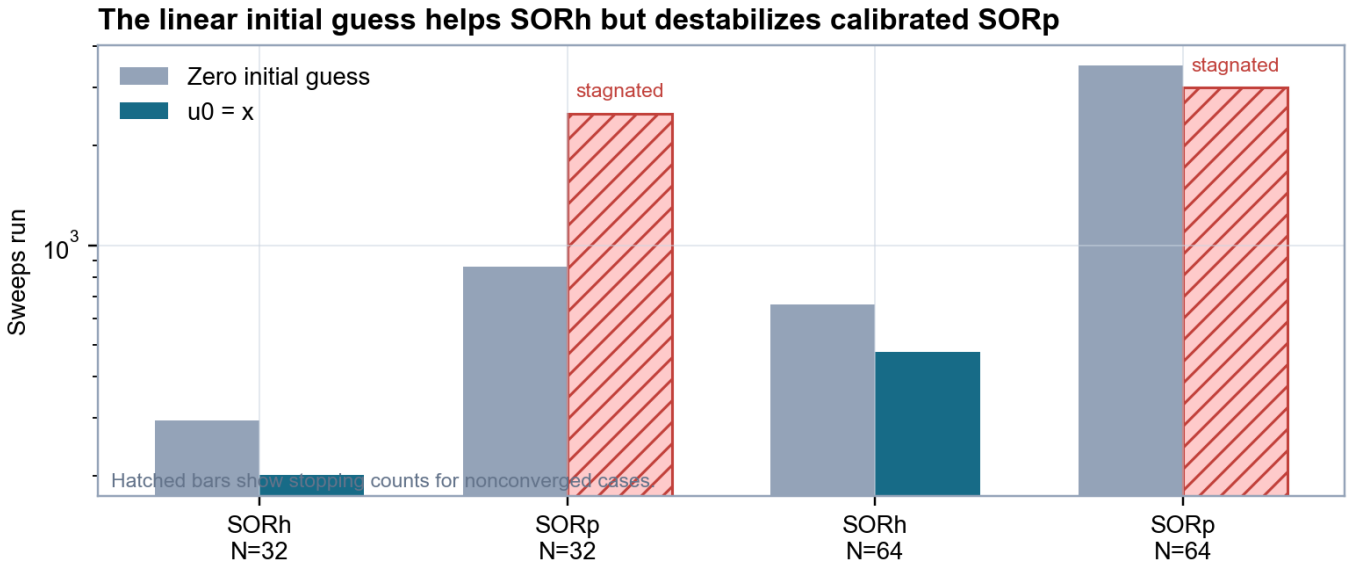



**Figure 6.** *SORh converges faster from u0=x. SORp does not reach tolerance before the stagnation safeguard; hatched bar heights are stopping counts, not converged sweep counts.*

- SORh improves from 295 to 201 sweeps at N=32 and from 662 to 475 at N=64, reductions of 31.9% and 28.2%.
- SORp stagnates at both grids. At N=32 the best residual is 5.04e-5, while at N=64 the best residual is 3.29e-4 and later residual growth is observed.
- The result shows that a nonlinear optimum can depend on the trajectory through state space. It does not establish that SORp cannot converge from u0=x; a separate omega calibration for that initial condition may recover convergence.

# 7. Optimized baseline performance and grid scaling

The final comparison runs each method with its experimentally calibrated omega on N=16, 32, and 64. Both use fixed timesteps, the zero initial guess, identical stopping tolerances, and the original degenerate coefficient.

| N | H sweeps | P sweeps | Sweep speedup | Time speedup | Solution difference |
|---|---|---|---|---|---|
| 16 | 139 | 221 | 1.59x | 1.58x | 7.51e-07 |
| 32 | 295 | 863 | 2.93x | 2.87x | 1.04e-06 |
| 64 | 662 | 3502 | 5.29x | 5.19x | 9.22e-07 |

**Table 6.** *Optimized SORh versus SORp performance on the degenerate baseline.*

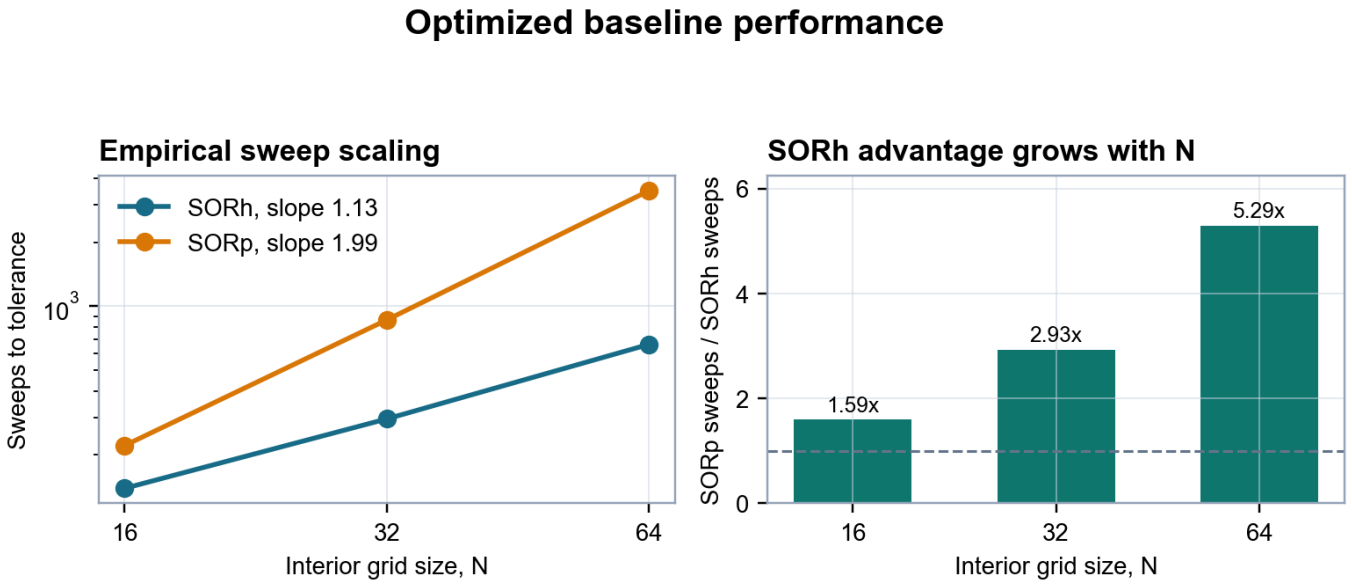



**Figure 7.** *Empirical sweep scaling from three grid sizes and the corresponding SORh speedup. The reported slopes are descriptive fits over N=16-64, not asymptotic proofs.*

- The empirical sweep exponents are 1.13 for SORh and 1.99 for SORp over the sampled range.
- The SORh sweep advantage grows from 1.59x at N=16 to 2.93x at N=32 and 5.29x at N=64. Elapsed-time speedups closely track the sweep ratios because the methods have nearly identical per-sweep cost.
- Normalized solution differences remain between 7.5e-7 and 1.0e-6, consistent with both methods converging to the same discrete solution at the requested tolerance.

# 8. Integrated interpretation

Taken together, the studies distinguish two regimes. In the constant-coefficient control, SORp behaves like the classical parabolic/SOR mechanism and is slightly faster. In the degenerate nonlinear regime, the useful SORp omega shifts toward a narrow stability edge and its sweep count grows nearly quadratically with N. SORh retains a near-formula omega, is insensitive to adaptive timestep changes, and scales much more favorably over the tested grids.

The H-CFL decomposition is especially important for the interpretation of SORh. Equalizing omega x CFL does not equalize convergence, so the update cannot be understood as only a residual correction with a combined relaxation coefficient. The independent directional role of omega is empirically large enough to control both stability and sweep count.

## Recommended baseline choices

- Use fixed timesteps for both methods on this problem. Adaptive timesteps add no SORh benefit and materially penalize SORp.
- Use the calibrated zero-state values: SORh omega=(0.8323, 0.8950, 0.9368) and SORp omega=(2.73, 2.68, 2.615) for N=(16, 32, 64).
- For changed initial guesses, coefficient families, orderings, or grid aspect ratios, recalibrate omega locally rather than assuming the baseline optimum transfers unchanged.
- Treat sweep count as the primary comparison metric. Record wall time as a secondary measure because these are single runs without repeated timing trials.

## Limitations

- The experiments cover one nonlinear PDE, one boundary condition, square grids, and lexicographic ordering. They do not establish universal SORh superiority.
- Each configuration is deterministic and was run once. There are no uncertainty bars for wall-clock time, although sweep counts are reproducible for a fixed configuration.
- The scaling exponents use only three grids. Larger grids are needed to claim an asymptotic rate.
- The degeneracy study supports a directional-propagation explanation through controlled correlations, but a mode-by-mode nonlinear analysis would be needed for a mechanistic proof.

## Reproducibility and data sources

| Study | Result directory |
|---|---|
| Omega calibration | 2D/results/nonlinear_studies/omega/20260803-040244 |
| Timestep | 2D/results/nonlinear_studies/timestep/20260803-055929 |
| H-CFL | 2D/results/nonlinear_studies/h_cfl/20260803-060108 |
| Degeneracy | 2D/results/nonlinear_studies/degeneracy/20260803-061736 |
| Initial guess | 2D/results/nonlinear_studies/initial/20260803-063356 |
| Performance | 2D/results/nonlinear_studies/performance/20260803-064210 |

All studies use the consolidated omega calibration automatically. The saved all_runs.csv files retain converged and nonconverged candidates, while the comparison files contain the selected first-branch optima or direct study comparisons.